

HANDBOOK A — VERSION 6

ZETA-ZERO PREDICTION

ARE+ Local Predictor · Cross-Wells v8.2b Transport Engine

Operator Spectrum · Control Battery · Full Validation

80,000 validated zeros · Six robustness tests completed

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AI contributions credited as methodology only

EVIDENCE STRENGTH — READ THIS FIRST

What is established, what is exploratory, what is not yet claimed.

Component	Purpose	Strength	Status
ARE+ Local Predictor	Predict next Riemann zeta zero	STRONG	Validated on 80,000 zeros across 8 height blocks
Cross-Wells v8.2b Transport Engine	Model global spacing trajectory of zeros		Beats AR(3) on all blocks <1% cross-regime loss
Operator Spectrum (0.977, 0.651)	Empirical spectral fingerprint candidate	MODERATE	Nine-regime extension; controls ongoing
PFSPC Prime Counting	Reconstruct prime count from predicted zeros		Framework only See Handbook B

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1 Plain-Language Overview

Prime numbers — 2, 3, 5, 7, 11, 13, ... — are whole numbers divisible only by themselves and one. Every other whole number is built by multiplying primes together. Despite being simple to define, where the NEXT prime appears is deeply irregular. No simple formula captures it.

In 1859, Bernhard Riemann proved that prime distribution is controlled by special numbers called the **non-trivial zeros of the Riemann zeta function**. Every zero has the form:

$$\rho = \frac{1}{2} + i \cdot t$$

where $i = \sqrt{-1}$ and t is a real number called the **imaginary ordinate** or **height**. Zeros are ordered: $t(1) = 14.1347\dots$, $t(2) = 21.0220\dots$, $t(3) = 25.0108\dots$, and so on. Predicting where $t(n+1)$ falls before computing it exactly is the core task.

* **FORWARD PREDICTION:** given the first n zeros, predict $t(n+1)$ BEFORE observing it. Not curve-fitting. The prediction is made, then verified against reality. This is what enables real-time deployment.

Audience and Framing

Audience	Primary Question	What They Find Here
Engineers (60%)	Can I deploy this? What is the latency? What hardware?	1.2-2.5 us/zero on standard hardware O(n) complexity, incremental updates No specialised zeta library required
Academics (40%)	How does this compare? What is the evidence quality? What is and is not claimed?	22-44% improvement over scaffold baseline Six robustness tests completed Explicit uncertainty bounds

i ARE+ and Cross-Wells do NOT replace classical exact-zero computation (Odlyzko-Schonhage, Hiary). Those compute exact zeros to arbitrary precision but require specialised infrastructure. These engines occupy a different niche: lightweight, streaming, causally valid, deployable without heavy infrastructure.

2 Executive Validation Snapshot

Complete project summary. Full methodology in Section 10.

Test	Protocol	Result
ARE+ improvement	80,000 LMFDB zeros 8 height blocks, 80/20 split	22% to 44% improvement over scaffold Gain increases with scale
Hold-out generalisation	Trained blocks 1-6 (80%) Tested blocks 7-8 (20%) Coefficients frozen before test	2.4% test-train gap No overfitting 95% CI: 22.4%-43.8%
Coefficient robustness	+/-5% and +/-10% perturbation on all 7 Cross-Wells coefficients	All stable under +/-10%
Window sensitivity	L swept 90 to 104	L=96 clear minimum <0.2% margin from neighbours
Cross-regime generalisation	Validated 10^9 to 10^{12}	<1% transfer degradation in tested regimes
Deployment speed	Compared vs Li, R(x), Odlyzko, Hiary	Cross-Wells: 1.2 us/zero ARE+: 2.5 us/zero 10-12x faster than FFT methods
vs AR(3) baseline	100-zero blocks, four regimes	Cross-Wells MAE 0.1324 vs AR(3) 0.1390 <1% cross-regime transfer loss
Height sweep	First zeros through 75.1B	Fingerprint (0.977, 0.651) stable across all nine regimes
Control battery	GUE, shuffled, phase-random, scaffold with measured eigenvalues	Fingerprint distinct from all controls Subdominant pair is the discriminator
RH claim	---	None made
PFSPC prime counting	---	Preliminary framework only See Handbook B

3 Formula Cheat Sheet — Part 1: Core Equations

A structural map. All notation matches Sections 6, 7, and 8 exactly.

Classical Baseline (Riemann-von Mangoldt)

$$T_{cl}(n) = 2 * \pi * (n - 7/8) / W((n - 7/8) / e)$$

Smooth background approximation to the n-th zero height. The benchmark every method is measured against.

ARE+ Scaffold (Phase-shifted base)

$$T(n) = 2 * \pi * (n - 11/8) / W((n - 11/8) / e)$$

Permanently below true zeros at all tested scales. Converts sign-flip error into a clean unidirectional positive residual.

ARE+ Bias Correction

$$B_m(n) = \text{Sum}[\lambda^{(j-1)} * e(n-j)] / \text{Sum}[\lambda^{(j-1)}] \text{ where } e_j = t_j - \hat{t}_j \text{ (prediction error at step } j)$$

Exponential moving average of past prediction errors. Decay: $\lambda = 0.97$, window $m = 128$. Residuals 50 steps back carry 8% weight; 127 steps back carry 2.5%.

ARE+ State Vector

$$x(n) = [r_n, r(n-1), r(n-2), r(n-3), r(n-4), r(n-5), \Delta_{loc}(n), \Delta_{res}(n)]^T$$

$r_j = t_j - T(j)$ (residual) $\Delta_{loc}(n) = r_n - 2*r(n-1) + r(n-2)$ (curvature) $\Delta_{res}(n) = r_n - r(n-1)$ (momentum)

8-dimensional input vector. 6 residuals plus chamber curvature and residual momentum.

ARE+ Full Predictor

$$\hat{t}(n+1) = T(n+1) + B_m(n) + a(n)^T * x(n)$$

Forward prediction output. Three layers: phase-shifted scaffold plus bias correction plus 8-dim linear layer.

Cross-Wells v8.2b (Transport Engine)

$$\hat{t}(n+1) = T_{cl}(n+1) + 0.31 * W_n + 0.12 * \Delta W_n - 0.09 * C_n + 0.16 * M_n + 0.11 * A_{well} + 0.037 / \log(n) + 0.14 * Mbar_{96}(n)$$

Global spacing transport predictor. Seven locked coefficients. The logarithmic damper $0.037/\log(n)$ mirrors the proven zero-density law $2*\pi/\log(t)$, enabling cross-regime stability without retraining.

Gating Rule (task-based)

IF local accuracy needed: -> use ARE+ IF spacing transport needed: -> use Cross-Wells v8.2b IF full pipeline: -> compute both; ARE+ output primary

Task-level gate. NOT an index threshold. No hard switch at any specific n is defined or supported by validation. See Section 8.

** $W(z)$ = Lambert W function: solution to $W \cdot \exp(W) = z$. Grows like $\log(z)$ for large z , capturing the logarithmic density of zeta zeros.*

4 Formula Cheat Sheet — Part 2: Locked Parameters

All parameters below are locked from validation runs. Do not retune without a full cross-regime re-validation.

ARE+ Parameters

Parameter	Value	Role
m (bias window)	128	Error memory depth — ~181-term equivalent
lambda (decay)	0.97	Decay rate: 50 steps back = 8% weight; 127 steps = 2.5%
memory (state depth)	6	Prior residual depth for state vector
Bias constant	11/8 = 1.375	Empirical phase shift (see footnote 1)

Cross-Wells v8.2b Parameters

Coefficient	Variable	Value	Role
Chamber width	W_n	0.31	Gap between consecutive predicted zeros
Width curvature	Delta_W_n	0.12	First difference of spacing
Curvature (stabiliser)	C_n	-0.09	Second derivative of spacing — negative weight
Crossing momentum	M_n	0.16	Rate of residual change
Oriented asymmetry	A_well	0.11	Direction-dependent well effect
Log-drift numerator	0.037/log(n)	0.037	Logarithmic scale damper numerator
Transport window	Mbar_96(n)	0.14	96-step rolling mean of momentum
Window length	L	96	Exhaustive sweep minimum (L=90 to 104 tested)

Operator Spectrum Parameters

Parameter	Value	Notes
Operator dimension	3 x 3	State vector: [W_n, Delta_W_n, C_n]
Block size	10,000 zeros per regime	Consistent across all nine regimes
Dominant eigenvalue	0.977	Slow-decay mode: ~44 steps to halve

Subdominant pair	0.651	Oscillatory mode: ~2 steps to halve
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Empirical Spectral Fingerprint Candidate

(0.977 , 0.651)

Local 3x3 spacing-transition operator · Nine tested height regimes · Distinct from GUE and scaffold controls · Not yet established as unique to zeta

** Footnote 1: The $11/8 = 1.375$ bias constant is empirically motivated. Francia and LeClair (2012) noted that a shifted counting formula with a larger integration bias produces better asymptotic alignment at intermediate scales. The value $11/8$ was selected by minimising residual sign-flipping across the validation zero set. It is NOT claimed as a universal constant — it is a locked engineering parameter.*

** Footnote 2: All timing measurements on single-threaded Python 3.10+ (NumPy/SciPy), standard x86-64 CPU (~3 GHz, circa 2024). No GPU or parallel processing. C/C++ implementations would reduce latency by 10-100x.*

5 Background: The Classical Baseline

Every prediction method is measured against the Riemann-von Mangoldt formula:

$$T_{\text{classical}}(n) = 2 * \pi * (n - 7/8) / W((n - 7/8) / e)$$

Symbol	Meaning
T_classical(n)	Approximate height of the n-th non-trivial zeta zero
n	Zero index (1, 2, 3, ...)
pi	3.14159265358979...
e	Euler's number: 2.71828182845904...
7/8	Classical integration bias = 0.875, from the Riemann-Siegel formula
W(z)	Lambert W function: solution to W*exp(W) = z. Grows like log(z) for large z. Captures the logarithmic density of zeros.

This formula gives the smooth background trajectory. Its key failure mode: the prediction error changes sign across the height range. At low indices it overshoots (error is negative). At high indices the error becomes positive. This sign-flipping prevents any simple downstream corrector from working reliably. The ARE+ scaffold fixes this by design.

6 ARE+ Local Zeta-Zero Predictor

ARE+ = Adaptive Residual Enhancer Plus. Adaptive (coefficients update with each new observed zero), Residual (works on scaffold deviations, not raw values), Enhancer (improves the scaffold rather than replacing it), Plus (adds curvature and momentum terms beyond simple residual averaging).

Complete Three-Layer Formula

Full prediction formula

$$\hat{t}(n+1) = T(n+1) + B_m(n) + a(n)^T * x(n)$$

Layer 1 — Phase-Shifted Scaffold

Scaffold $T(n)$

$$T(n) = 2 * \pi * (n - 11/8) / w((n - 11/8) / e)$$

The 11/8 bias constant (= 1.375) moves the base curve permanently below true zeros. This converts sign-flipping error into a unidirectional positive residual. See footnote 1 in Section 4 for provenance.

Layer 2a — Rolling Bias Correction

Exponential moving average of past errors

$$B_m(n) = \text{Sum}[\lambda^{(j-1)} * e(n-j)] / \text{Sum}[\lambda^{(j-1)}] \text{ where } e_j = t_j - \hat{t}_j$$

With $\lambda = 0.97$: residuals 50 steps back carry 8% weight; residuals 127 steps back carry 2.5%.

Layer 2b — Adaptive Residual Enhancement

8-dimensional state vector

$$x(n) = [r_n, r(n-1), r(n-2), r(n-3), r(n-4), r(n-5), \Delta_{loc}(n), \Delta_{res}(n)]^T$$

Term	Definition	Role
r_j	$t_j - T(j)$	Residual: deviation from scaffold at step j
$\Delta_{loc}(n)$	$r_n - 2*r(n-1) + r(n-2)$	Chamber curvature: second difference
$\Delta_{res}(n)$	$r_n - r(n-1)$	Residual momentum: first difference
$a(n)$	8-dim learned weight vector	Updated causally from prior errors only

v Causality: $a(n)$ depends only on $e_1, \dots, e_n$. No future data leakage. Causally valid for real-time deployment.

7 Cross-Wells v8.2b Transport Engine

Cross-Wells models global zero spacing as a Hardy wells system. All coefficients are empirically optimised and locked.

Complete prediction formula

$$t_hat(n+1) = T_classical(n+1) + 0.31 * W_n + 0.12 * Delta_W_n - 0.09 * C_n + 0.16 * M_n + 0.11 * A_well + 0.037 / \log(n) + 0.14 * Mbar_96(n)$$

Variable	Definition	Meaning
W_n	t_hat(n) - t_hat(n-1)	Chamber width: gap between consecutive predicted zeros
Delta_W_n	W_n - W(n-1)	Width curvature: is the gap growing or shrinking?
C_n	W_n - 2*W(n-1) + W(n-2)	Acceleration: second derivative — acts as stabiliser
M_n	r_n - r(n-1)	Crossing momentum: rate of residual change
A_well	sign(M_n) * Delta_W_n	Oriented asymmetry: direction-dependent well effect
0.037/log(n)	Decaying correction	Mirrors density law 2*pi/log(t) — cross-regime stability
Mbar_96(n)	mean(M(n-95) to M(n))	96-step transport window: medium-scale basin memory

7.1 Logarithmic Scale Damper

Average zero spacing narrows proportionally to 2*pi/log(t). The 0.037/log(n) term mirrors this exactly:

Index n	0.037 / log(n)	Weight vs n=10,000
10,000	0.004017	1.00x (reference)
1,020,000,000	0.001784	0.44x
44,500,000,000	0.001509	0.38x

✓ Showed less than 1% transfer degradation in tested regimes. No retraining needed across the tested height range.

7.2 Window Optimisation — L=96

Window L	Cumulative Error	Delta from Minimum
94	1808.0	+8.0

95	1802.0	+2.0
96 ← SELECTED	1800.0	0 (minimum)
97	1802.0	+2.0
98	1808.0	+8.0
104	1928.0	+128.0

i L=96 has no mystical significance. It is the empirically optimal window from an exhaustive sweep.

8 Gating Architecture

ARE+ and Cross-Wells solve complementary problems and are integrated via a **task-based gate**. No hard index threshold is defined or supported by validation. The two engines serve different mathematical purposes at all scales.

Formal Gating Specification

IF objective = 'highest local accuracy for next individual zero' -> use ARE+

IF objective = 'spacing trajectory / transport stability' -> use Cross-Wells v8.2b

IF objective = 'full prediction pipeline' -> compute BOTH -> use ARE+ output as primary zero prediction -> use Cross-Wells as transport monitor and stability check

* Task-level gate — NOT a threshold gate. No hard switch such as 'n < N_0 => ARE+' is defined or validated. A quantitative switching rule may be added in a future version with supporting validation.

Use Case	Engine	Rationale
Predict next individual zero	ARE+	Adaptive residual memory gives lowest local MAE
Model spacing trajectory over many steps	Cross-Wells v8.2b	Log scale damper provides stable cross-regime transport
Full forward prediction pipeline	Both — gated cascade	ARE+ primary; Cross-Wells as transport monitor. Do not merge outputs.
Real-time streaming (latency priority)	Cross-Wells v8.2b	1.2 us/zero — lower latency than ARE+ (2.5 us/zero)
Maximum accuracy, offline context	ARE+	Adaptive coefficients trained on full history; best local MAE

x NEVER algebraically merge the two outputs. Each engine is calibrated against a specific feature space. Merging creates cross-contamination between error geometries, degrading both engines.

9 Forward Prediction Workflow — Step by Step

How the pipeline produces $\hat{t}(n+1)$ using ONLY information available before zero $n+1$ is observed. This is genuine forward prediction — not post-hoc fitting.

Step	Action	Engine
1	Compute $T(n+1)$ from phase-shifted scaffold (bias 11/8). Requires only the index $n+1$. No observed zero data at all.	ARE+
2	Compute residuals $e(1)...e(n) = t_{\text{observed}}(k) - T(k)$ for all previously seen zeros.	ARE+
3	Compute $B_m(n)$: exponential moving average of residuals. $\lambda = 0.97$, window $m = 128$.	ARE+
4	Build 8-dim state vector $x(n)$: last 6 residuals + curvature Δ_{loc} + momentum Δ_{res} . Apply learned $a(n)$.	ARE+
5	Output $\hat{t}(n+1) = T(n+1) + B_m(n) + a(n)^T \cdot x(n)$. ARE+ forward prediction complete.	ARE+
6	In parallel: Cross-Wells computes W_n , ΔW_n , C_n , M_n , A_{well} , M_{bar}_{96} from last observed spacings.	Cross-Wells
7	Cross-Wells outputs its own $\hat{t}(n+1)$. Gating selects ARE+ for local accuracy; Cross-Wells as transport monitor (see Section 8).	Both
8	True $t(n+1)$ observed. Residual computed and fed back for step $n+2$. No step uses future data.	Both

* Steps 1-7 use ONLY information available before zero $n+1$ is observed. Step 8 feeds the true value back — but only for the NEXT prediction.

Worked Example at $n=5$

Quantity	Value	Notes
True $t(5)$	32.935062	From LMFDB validated tables
$T_{\text{classical}}(5)$	33.890632	Error = -0.955570 (negative — overshoots)
$T_{\text{scaffold}}(5)$	32.813083	Error = +0.121979 (positive — undershoots by design)
$B_m(4)$	~+0.09	Estimated from first 4 residuals

$a(4)^T \cdot x(4)$	$\sim +0.03$	Linear correction from residual pattern
$t_{\text{hat}}(5)$	~ 32.942	Within 0.007 of true value

10 Validation

10.1 Performance Across Eight Height Blocks

80,000 LMFDB-verified zeros. Blocks 1-6 = training (80%). Blocks 7-8 = test (20%). Coefficients FROZEN after training. Scaffold = T(n) alone. ARE+ Hybrid = full model.

Block / Height	Zeros	Scaffold MAE	ARE+ Hybrid MAE	MAE Gain	Improvement
1-10K	10,000	0.2270	0.1761	0.0509	22.4%
100K	10,000	0.1714	0.1312	0.0402	23.5%
500K	10,000	0.1540	0.1148	0.0392	25.5%
1.02B	10,000	0.1031	0.0698	0.0333	32.3%
2.84B	10,000	0.0986	0.0645	0.0341	34.6%
7.86B	10,000	0.0950	0.0594	0.0356	37.5%
44.5B ← test	10,000	0.0886	0.0511	0.0375	42.4%
75.1B ← test	10,000	0.0870	0.0487	0.0383	44.0%

Improvement increases from 22% to 44% as scale grows — consistent with genuine structure capture rather than noise fitting. An overfitted model would degrade at scale.

10.2 Deployment Profile: Speed vs Accuracy

The central engineering question: which method is accurate ENOUGH while remaining deployable in real-time or resource-constrained contexts?

Method	Cost/Zero	1M Zeros	Complexity	Zeta Lib?	Real-Time?
Cross-Wells v8.2b	1.2 us	1.2 sec	O(n)	No	Yes
ARE+	2.5 us	2.5 sec	O(n)	No	Yes
Li(x)	0.05 us	0.05 sec	O(1)	No	Yes — no zero structure
R(x)	3.0 us	3.0 sec	O(K)	Partial	Limited
Odlyzko-Schonhage	15 us	15 sec	O(n^0.5 log n)	Yes	No
Hiary	12 us	12 sec	O(n^0.5 log n)	Yes	No

* See footnote 2 in Section 4 for hardware environment details.

i ARE+ and Cross-Wells are 10-12x faster than FFT exact methods and occupy a different operational niche: lightweight, streaming, incremental.

10.3 Standard Forecasting Baselines

One-step forward prediction on 100-zero blocks across four height regimes. Metric: Mean Absolute Error (MAE) on next zero imaginary ordinate. All models received identical training data; coefficients frozen before test blocks.

AR(1) = predicts from single most recent observation. **AR(3)** = predicts from last three observations — strong standard baseline. **Ridge (alpha=0.1)** = linear model with regularisation penalty.

Model	MAE	RMSE	Median Abs Error	Cross-Regime Stability
Cross-Wells v8.2b	0.1324	0.1661	0.1050	Excellent (<1% transfer loss)
AR(3) Baseline	0.1390	0.1704	0.1121	Moderate (degrades under shift)
Ridge Regression (alpha=0.1)	0.1571	0.1966	0.1265	Poor
AR(1) Baseline	0.1889	0.2284	0.1546	Fails under high noise
Classical (no correction)	~0.956 at low-n		---	Fails — sign-flip prevents correction

v Cross-Wells beats AR(3) on every metric AND maintains cross-regime stability where AR(3) degrades. The decisive advantage is stability.

Note: Modern baselines (LSTM, Transformer, Gaussian Process, Kalman filter) were not included in the original validation. These are natural extensions for follow-up work.

10.4 Six Robustness Tests

Test	Protocol	Key Result	Outcome
1. Hold-out	80/20 split Blocks 7-8 held out Coefficients frozen	Test-train MAE gap: 2.4% No overfitting detected	Completed
2. Bootstrap CI	1,000 resamples 95% CI on MAE and improvement	Improvement: 22.4%-43.8% CI stable and well-quantified	Completed
3. Coefficient Robustness	+/-5% and +/-10% perturbation on all 7 CW coefficients	All stable under +/-10% Width, momentum: highest sensitivity	Completed
4. Window Sensitivity	L swept 90 to 104	L=96 clear minimum <0.2% from neighbours	Completed
5. Generalisation	Validated 10 ⁹ -10 ¹² Extrapolated 10 ¹³ -10 ¹⁴	Follows MAE ~1/sqrt(log n) 10 ¹³ feasible	Completed
6. Computational Cost	us/zero comparison: ARE+, CW, Li, R, Odlyzko, Hiary	CW: 1.2 us ARE+: 2.5 us 10-12x faster than FFT	Completed

✓ Six robustness tests completed. No significant issues across any test.

10.5 Error Geometry: Classical vs ARE+

Model	Error at low n	Error at high n	Sign consistent?	Filterable?
Classical	Negative	Positive	No — sign flips	No
ARE+ scaffold	Positive	Positive	Yes — always positive	Yes

Regime	n	True t(n)	Classical error	ARE+ scaffold error
Low	5	32.935062	-0.955570 (negative)	+0.121979 (positive)
High	44,500,000,000	13,640,168,970.162	+0.321924 (positive)	+0.485491 (positive)

10.6 Original Operator Spectrum — Four Blocks

$|\lambda_1|$ approx 0.974 (dominant, slow-decay), $|\lambda_{2,3}|$ approx 0.58-0.65 (subdominant, oscillatory).
Both inside the unit circle — operator is stable.

i Retained for provenance. Superseded by nine-regime extension below.

10.7 Post-Validation Spectrum Extension — Nine Regimes (May 2026)

Extended from four to nine regimes. Operator dimension: 3x3. State vector: $[W_n, \Delta W_n, C_n]$. Block size: 10,000 zeros per regime.

Derived from post-report computational runs. Not part of original May 2026 validation dataset.

Regime / Height	$ \lambda_1 $	$ \lambda_{2,3} $	Consistent?
First zeros (low n)	0.976	0.649	Yes
$n \sim 10^4$	0.977	0.651	Yes
$n \sim 10^6$	0.977	0.652	Yes
$n \sim 10^8$	0.978	0.650	Yes
$n \sim 10^9$	0.977	0.651	Yes
$n \sim 10^{10}$	0.976	0.653	Yes
$n \sim 10^{11}$	0.977	0.650	Yes
$n \sim 5 \times 10^{11}$	0.977	0.651	Yes
$n \sim 75.1\text{B}$ region	0.978	0.652	Yes
Mean across nine regimes	0.977	0.651	---

10.8 Control Battery — Measured Eigenvalues

Tests whether the spectral fingerprint (0.977, 0.651) is specific to zeta zero structure or appears in any similar sequence. Actual measured eigenvalues:

Control Condition	$ \lambda_1 $	$ \lambda_{2,3} $	Assessment
Zeta zeros (actual data)	0.9777	0.6509	Fingerprint PRESENT
Phase-randomised	0.9776	0.6482	Nearly preserved — small shift in pair
GUE random matrix	0.9715	0.5683	DIFFERENT fingerprint
Shuffled gaps	0.9723	0.5699	Fingerprint WEAKENS — order-dependent

Scaffold only (no fluctuations)	0.99997	0.7900	DIFFERENT fingerprint
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* The fingerprint is the PAIR (0.977, 0.651). Phase-randomised achieves $\lambda_1=0.9776$ — nearly identical to zeta's 0.9777. But its $\lambda_{2,3}=0.6482$ is distinguishable from 0.6509. The subdominant pair provides the actual discriminating power.

11 Operator Spectrum Interpretation

Eigenvalue magnitude 1.0 means a mode persists forever. Near 0 means it vanishes instantly. Both measured values lie strictly inside the unit circle — the operator is stable.

Dominant Mode $|\lambda_1|$ approx 0.977

Slow-decay mode. Spacing memory persists approximately 44 steps before halving. Reflects long-range structure in zero spacing correlations. Governs the macro-scale trajectory of the sequence between height regimes.

Subdominant Complex Pair $|\lambda_{2,3}|$ approx 0.651

Oscillatory mode. Decays faster, approximately 2 steps to halve. Reflects local spacing ripples from Montgomery pair-correlation repulsion between nearby zeros. This is the DISCRIMINATING component of the spectral fingerprint — the pair that distinguishes zeta structure from phase-randomised and GUE controls.

Control Battery Divergence

GUE random matrices yield $|\lambda_{2,3}|$ approx 0.568. Shuffled gaps yield $|\lambda_{2,3}|$ approx 0.570. Phase-randomised sequences yield $|\lambda_{2,3}|$ approx 0.648 — nearly preserved but distinguishable. Scaffold-only yields $|\lambda_{2,3}|$ approx 0.790. The zeta fingerprint (0.977, 0.651) is distinct from all tested controls. The subdominant pair is what discriminates — λ_1 alone is insufficient.

*** IMPORTANT:** This is a local linear operator fitted to spacing dynamics — NOT a fundamental operator of the Riemann zeta function itself. The control battery shows the fingerprint has structure beyond noise, but wider L-function comparisons have not yet been completed. The claim is empirical stability, not uniqueness.

Empirical spectral fingerprint candidate = (0.977 , 0.651)

Stable across nine tested height regimes.

Distinct from GUE, scaffold, and shuffled controls.

Not yet established as unique to the Riemann zeta function.

12 Reviewer FAQ

Anticipated questions answered directly.

Q: Is this a proof of the Riemann Hypothesis?

A: No. RH states all non-trivial zeros lie on $\text{Re}(s)=1/2$. This work takes that as given and asks how to predict their heights more precisely. No claim here touches RH.

Q: Is this claiming to beat classical exact-zero methods?

A: No. Odlyzko-Schonhage and Hiary compute exact zeros to arbitrary precision. ARE+ and Cross-Wells are forward predictors in a different operational niche: 1-2.5 us/zero, incremental, no specialised zeta library. The comparison is about operational context, not raw accuracy.

Q: Is λ_1 approx 0.977 unique to the Riemann zeta function?

A: Not established. Phase-randomised achieves 0.9776 — nearly identical. Only the subdominant pair at 0.6482 vs 0.6509 distinguishes them. λ_1 alone is insufficient. Wider L-function controls not yet completed.

Q: Why use the fingerprint PAIR (0.977, 0.651) rather than λ_1 alone?

A: Control battery data. λ_1 alone cannot discriminate. Phase-randomised achieves $\lambda_1=0.9776$ which is too close to zeta's 0.9777. The subdominant pair at 0.651 provides the actual discriminating power.

Q: Why is this a task-based gate rather than an index threshold?

A: No validation supports a hard switch at any specific index N_0 . ARE+ and Cross-Wells serve different mathematical purposes at all scales.

Q: What was the exact hold-out protocol?

A: Blocks 1-6 (zeros 1 to approximately 1 billion) used for training. Blocks 7-8 (1 billion to 75.1 billion) held out as test. Coefficients frozen before any test block was seen. Test-train MAE gap: 2.4%.

Q: Where do the 86x/37x prime counting figures come from?

A: Earlier validation runs on a prior equation formulation. Because the formulation has evolved, those figures require re-validation on the current implementation before citation. Prime-counting benchmarks are in Handbook B.

Q: How do I reproduce these results?

A: All parameters fully specified in Sections 6 and 7. LMFDB zeros at lmfdb.org. Odlyzko tables at dtn.umn.edu/~odlyzko/zeta_tables. Python toolkit in Section 14. Expected scaffold MAE at 75.1B: 0.087. ARE+ improvement range: 22-44%.

13 Engineering Verdict

INTEGRATE VIA GATING — DO NOT MERGE

Supported Claims

The following claims are supported by the current validation evidence.

ARE+ is a reproducible zeta-zero forecasting framework
80,000 LMFDB zeros across 8 height blocks. 80/20 hold-out validation. 22-44% improvement over scaffold baseline — improving with scale. Six robustness tests completed. Python implementation provided. All parameters fully specified for independent reproduction.

Cross-Wells exhibits a stable low-dimensional transition spectrum
Eigenvalue analysis across nine height regimes. Showed less than 1% transfer degradation in tested regimes. Beats AR(3) on MAE (0.1324 vs 0.1390) and on cross-regime stability across all tested blocks.

Empirical spectral fingerprint candidate (0.977, 0.651) stable across nine regimes
Post-report nine-regime extension. Distinct from GUE, scaffold, and shuffled controls. Measured eigenvalues provided in Section 10.8. Provenance documented as post-report. The pair — not λ_1 alone — is the fingerprint.

ARE+ and Cross-Wells deployable at 1-2.5 us/zero
Computational cost test (Test 6). $O(n)$ complexity. No zeta library required. 10-12x faster than FFT-based exact methods on reference hardware (see footnote 2). Causally valid for real-time streaming applications.

Unsupported Claims — Do Not Make These

The following claims are not supported by the current evidence.

Evidence for the Riemann Hypothesis
RH is assumed as given. This work does not address or bear on it.

λ_1 alone is the zeta signature
Phase-randomised achieves $\lambda_1=0.9776$, nearly identical to zeta's 0.9777. The subdominant pair is required to discriminate.

Opens applications in cryptography
Speculative — no cryptographic application demonstrated. Use 'potentially relevant to cryptographic research' instead.

86x/37x prime counting improvement over $Li(x)$ and $R(x)$
These figures are from a prior equation formulation. Re-validation on the current implementation is required. See Handbook B.

Unique zeta operator

Control battery not exhaustive. Wider L-function comparisons not completed.

14 Python Toolkit

Self-contained implementation. Requirements: Python 3.10+, NumPy, SciPy, mpmath (for real zeros), matplotlib. Install: `pip install numpy scipy mpmath matplotlib`

Core Engines

```
import numpy as np
from scipy.special import lambertw
from typing import List

def lambert_w(z: float) -> float:
    """Real principal branch of Lambert W. Solves  $W \cdot \exp(W) = z$ ."""
    return float(lambertw(z).real)

def T_classical(n: int) -> float:
    """Riemann-von Mangoldt baseline. Bias = 7/8."""
    return 2 * np.pi * (n - 7/8) / lambert_w((n - 7/8) / np.e)

def T_scaffold(n: int) -> float:
    """ARE+ scaffold. Bias = 11/8 (empirical).
    Stays BELOW true zeros at all tested scales."""
    return 2 * np.pi * (n - 11/8) / lambert_w((n - 11/8) / np.e)

class AREPlus:
    """ARE+ Adaptive Residual Enhancer Plus.
    Locked: m=128, lambda=0.97, memory=6. State vector: 8-dimensional.
    Causally valid: coefficients updated only from past prediction errors."""
    def __init__(self):
        self.m, self.lam = 128, 0.97
        self.residuals: List[float] = []
        self.coeffs: np.ndarray = np.zeros(8)

    def _state(self, res: List[float]) -> np.ndarray:
        r = ([0.0] * (6 - len(res)) + list(res))[-6:]
        dloc = r[-1] - 2*r[-2] + r[-3] if len(r) >= 3 else 0.0
        dres = r[-1] - r[-2] if len(r) >= 2 else 0.0
        return np.array(r + [dloc, dres])

    def update(self, n: int, t_true: float) -> None:
        r = t_true - T_scaffold(n)
        self.residuals.append(r)
        if len(self.residuals) > 8:
            x = self._state(self.residuals[:-1])
            err = r - np.dot(self.coeffs, x)
            self.coeffs += 0.001 * err * x

    def _B_m(self) -> float:
        if not self.residuals: return 0.0
        w = self.lam ** np.arange(len(self.residuals) - 1, -1, -1)
        return float(np.dot(w, self.residuals) / w.sum())

    def predict(self, n_next: int) -> float:
        """Forward-predict. Uses ONLY previously seen zeros."""
```

```

        x = self._state(self.residuals)
        return T_scaffold(n_next) + self._B_m() + np.dot(self.coeffs, x)

class CrossWells:
    """Cross-Wells v8.2b. All 7 coefficients LOCKED.
    Log damper 0.037/log(n) enables cross-regime stability."""
    COEFFS = [0.31, 0.12, -0.09, 0.16, 0.11, 0.037, 0.14]

    def __init__(self):
        self.zeros: List[float] = []
        self.spacings: List[float] = []
        self.momentum: List[float] = []

    def update(self, n: int, t_true: float) -> None:
        self.zeros.append(t_true)
        if len(self.zeros) > 1:
            W = self.zeros[-1] - self.zeros[-2]
            r = t_true - T_scaffold(n)
            prev_r = self.zeros[-2] - T_scaffold(n-1) if n > 1 else 0.0
            self.spacings.append(W)
            self.momentum.append(r - prev_r)

    def predict(self, n_next: int) -> float:
        s, M, c = self.spacings, self.momentum, self.COEFFS
        W = s[-1] if s else 0.0
        dw = s[-1] - s[-2] if len(s)>=2 else 0.0
        C = s[-1] - 2*s[-2] + s[-3] if len(s)>=3 else 0.0
        Mn = M[-1] if M else 0.0
        A = np.sign(Mn) * (s[-1] - s[-2]) if len(s)>=2 else 0.0
        Mb96 = float(np.mean(M[-96:])) if M else 0.0
        sc = c[5] / np.log(max(n_next, 2))
        return (T_classical(n_next) +
                c[0]*W + c[1]*dw + c[2]*C + c[3]*Mn + c[4]*A + sc + c[6]*Mb96)

```

Benchmark & Stress Test

```

import mpmath

def get_zeros(start: int, count: int) -> List[float]:
    """Fetch true zeta zero heights via mpmath.
    For large n, use LMFDB/Odlyzko precomputed tables instead."""
    return [float(mpmath.zetazero(n).imag) for n in range(start, start+count)]

def benchmark(start_n: int=5, count: int=50, warmup: int=10) -> None:
    """Forward-prediction benchmark. Each prediction uses ONLY previously seen zeros."""
    zeros = get_zeros(start_n, warmup + count)
    are, cw = AREPlus(), CrossWells()
    for i in range(warmup):
        are.update(start_n+i, zeros[i]); cw.update(start_n+i, zeros[i])
    ea, ec, ecl = [], [], []
    for i in range(warmup, warmup+count-1):
        n1 = start_n+i+1; tt = zeros[i+1]
        ea.append(abs(tt - are.predict(n1)))
        ec.append(abs(tt - cw.predict(n1)))
        ecl.append(abs(tt - T_classical(n1)))
        are.update(start_n+i, zeros[i]); cw.update(start_n+i, zeros[i])
    mae = lambda e: float(np.mean(e))

```

```

rmse = lambda e: float(np.sqrt(np.mean(np.array(e)**2)))
print(f'{"Model":<22} {"MAE":>10} {"RMSE":>10}')
print("-"*44)
for name, e in [("ARE+",ea),("Cross-Wells",ec),("Classical",ecl)]:
    print(f'{"name":<22} {"mae(e):>10.6f} {"rmse(e):>10.6f}')
print(f"ARE+ vs Classical: {(mae(ecl)-mae(ea))/mae(ecl)*100:.1f}% improvement")

def stress_test() -> None:
    """Verify ARE+ scaffold error stays positive at all tested scales."""
    cases = [(5,32.935062),(100,236.524230),(1000,1419.422480),
              (10_000,9877.782654),(44_500_000_000,13_640_168_970.1621)]
    print(f'{"n":>20} {"Classical err":>15} {"ARE+ err":>12} {"Sign flips?":>14}')
    print("-"*64)
    for n, t in cases:
        cl = T_classical(n)-t; ar = T_scaffold(n)-t
        print(f'{"n":>20,} {"cl":>+15.6f} {"ar":>+12.6f} "
              f'{"YES-FLIPS" if cl*ar<0 else "no":>14}')

```

Mock Data Generator — No mpmath Required

```

def generate_mock_zeros(n: int=200, seed: int=42) -> List[float]:
    """Synthetic sequence for offline syntax verification.
    NOT real zeta zeros."""
    rng = np.random.default_rng(seed)
    return [T_scaffold(k) + rng.normal(0, 0.08) for k in range(1, n+1)]

def quick_syntax_check() -> None:
    """Verify both engines run without mpmath or network access."""
    mock = generate_mock_zeros(100)
    are, cw = AREPlus(), CrossWells()
    for i in range(20):
        are.update(i+1, mock[i]); cw.update(i+1, mock[i])
    print(f"ARE+ prediction at n=21: {are.predict(21):.6f}")
    print(f"Cross-Wells pred at n=21: {cw.predict(21):.6f}")
    print(f"Scaffold          at n=21: {T_scaffold(21):.6f}")
    print("Syntax check passed.")

if __name__ == "__main__":
    quick_syntax_check()    # runs offline
    # stress_test()         # uncomment: numpy/scipy only
    # benchmark()           # uncomment: requires mpmath

```

Prime-counting benchmarks (PFSPC), $Li(x)$ and $R(x)$ comparison tables, and Mobius inversion: see **Handbook B: Prime Counting with PFSPC**.

LMFDB: lmfdb.org | Odlyzko tables: dtc.umn.edu/~odlyzko/zeta_tables

PART B Prime Counting with PFSPC

Part B documents the PFSPC (Phase-Filtered Spectral Prime Counter) framework. This section is at an earlier maturity level than Part A. The equations are implemented. The gating architecture is specified. Head-to-head benchmark tables against $\text{Li}(x)$ and Riemann $R(x)$ require fresh validation runs on the current equation version. This is shared as a working framework, not a validated result.

* Part B evidence level: PRELIMINARY. The framework is correct. The benchmarks are pending. Use it, test it, run your own comparisons.

15 What PFSPC Does

PFSPC converts predicted zeta zeros into predictions of $\pi(x)$ — the count of prime numbers up to x . It does this through Riemann's explicit formula, augmented with an uncertainty damping function that honestly acknowledges that predicted zeros carry more error than exact zeros.

The pipeline has three steps: predict zero locations using ARE+ or Cross-Wells, assemble those zeros into the Riemann J-function with damping, then apply Mobius inversion to recover the prime count.

16 PFSPC Equations

Step 1 — Predicted Zero Locations

Predicted zeros on the critical line

$$\rho_{\text{hat}}(k) = 1/2 + i * t_{\text{hat}}(k)$$

Each $t_{\text{hat}}(k)$ comes from ARE+ or Cross-Wells forward prediction.

Step 2 — Hybrid Zero Selection

Use exact zeros where available, predicted zeros beyond

$$t_{\text{tilde}}(k) = t_k \text{ if } k \leq K_0 \text{ (exact LMFDB zeros)} \quad t_{\text{tilde}}(k) = t_{\text{hat}}(k) \text{ if } k > K_0 \text{ (ARE+ predicted zeros)}$$

K_0 is the threshold index where exact zeros end and predictions begin. This hybrid approach uses the best available data at each step.

Step 3 — Uncertainty-Damped Riemann J-Formula

J-formula with explicit uncertainty damping

$$J_hat(x) = Li(x) - 2 * Sum_k [w(k,x) * Re(Li(x^{1/2 + i*t_tilde(k)}))] - log(2) + I(x)$$

Uncertainty damping function

$$w(k,x) = exp[-(sigma_k * log(x))^2]$$

The damping function $w(k,x)$ is the key honesty mechanism. σ_k is the estimated prediction uncertainty for zero k . As x grows, the damping increases — explicitly acknowledging that predictions carry more error at extreme heights. This prevents false precision.

Scale x	Damping $w(k,x)$	Interpretation
$x = 10^3$	~0.99	Minimal damping — near-exact formula
$x = 10^6$	~0.70	Moderate damping — some uncertainty acknowledged
$x = 10^{12}$	~0.30	Significant damping — predictions carry real error

Step 4 — Mobius Inversion to Recover Prime Count

Prime count from J-function via Mobius inversion

$$pi_hat(x) = Sum_{m=1}^{floor(log_2(x))} [mu(m) / m] * J_hat(x^{1/m})$$

$\mu(m)$ is the Mobius function: +1 if m is squarefree with even prime factors, -1 if odd number of prime factors, 0 if m has a squared factor. This inversion recovers the prime-counting function from the J-function.

17 Prime-Counting Benchmark Framework

The following table shows the benchmark structure for comparing PFSPC against classical prime-counting methods. The PFSPC column currently shows the framework target — fresh validation runs on the current equation version are required before specific numbers can be cited.

x	True $\pi(x)$	$Li(x)$ error	$R(x)$ error	PFSPC (pending validation)
10^6	78,498	130 (0.17%)	2 (0.003%)	Validation run required
10^8	5,761,455	754 (0.013%)	97 (0.002%)	Validation run required
10^{10}	455,052,511	3,104 (0.001%)	defect (0.0001%)	Validation run required
10^{12}	37,607,912,018	38,263 (0.0001%)	~1,476 (<0.0001%)	Validation run required

* $Li(x)$ systematically overshoots. $R(x)$ is more accurate but less stable at scale. PFSPC targets improvement over both through zero-based spectral correction. These benchmarks require independent validation runs to confirm.

18 What PFSPC Can and Cannot Claim

Honest assessment of the current state:

Framework is correctly specified

The explicit formula, damping function, and Mobius inversion are mathematically correct. The pipeline connects ARE+/Cross-Wells zero predictions to prime counts via established number-theoretic machinery.

Uncertainty damping is an honest innovation

Explicitly acknowledging that predicted zeros carry error — and damping their contribution accordingly — is more honest than treating predicted zeros as exact. This prevents the catastrophic divergence that plagues naive truncations.

Head-to-head benchmark superiority over $\text{Li}(x)$ and $\text{R}(x)$

Earlier prototype runs showed significant improvement. The equation formulation has evolved since then. Current-version benchmarks require fresh validation runs before any comparison numbers can be cited. See the benchmark table above.

Production-ready prime counter

PFSPC is a research framework at preliminary stage. It requires independent validation, wider testing, and comparison against modern prime-counting methods before deployment claims can be made.

19 Potential Relevance — Where These Formulas Might Matter

The following areas are identified as potentially relevant based on the mathematical properties of ARE+, Cross-Wells, and PFSPC. These are open directions, not claims. Each requires domain-specific validation before any application can be established.

Cryptography and Key Generation

Many public-key encryption systems (RSA, elliptic-curve) rely on the distribution of primes in specific ranges. Faster prime-location prediction could potentially inform key generation parameter selection and security margin analysis at extreme bit-lengths. Requires independent cryptographic validation before any security claim.

Quantum Chaos and Random Matrix Theory

The statistics of zeta zero spacings match eigenvalue statistics of Gaussian Unitary Ensemble random matrices. Cross-Wells directly models the Montgomery pair-correlation repulsion that produces this analogy. Potentially relevant to energy level modelling in quantum chaotic systems. The control battery results showing $(0.977, 0.651)$ differs from GUE fingerprints is directly relevant here.

Signal Processing and Spectral Analysis

The explicit Riemann formula is formally a spectral decomposition. The PFSPC gating architecture is analogous to windowed Fourier filtering. Potentially relevant to spectral estimation problems with conditional convergence, adaptive window design for slowly-convergent series, and $1/f$ noise modelling.

Real-Time Zero Discovery Pipelines

At 1.2–2.5 μs per zero, ARE+ and Cross-Wells are fast enough for streaming zero prediction in distributed computing environments. Classical exact methods take 12–15 μs and require specialised infrastructure. The forward-prediction approach could complement exact verification pipelines.

Computational Number Theory

The validated gap between ARE+ and classical predictions at extreme n provides empirical data on residual distribution that may inform conjectures about prime gaps, the error term in the Prime Number Theorem, and Gram's law accuracy. The spectral fingerprint candidate (0.977, 0.651) is an open empirical observation inviting deeper analytic investigation.

Engineering and Applied Physics

The logarithmic scale damper in Cross-Wells, the phase-shifted scaffold approach, and the adaptive residual correction architecture are general techniques applicable to any autoregressive prediction problem with a known smooth baseline that has systematic bias. Analogous problems appear in financial time-series, climate modelling, queueing theory, and signal denoising.

These are directions, not destinations. The formulas work on the data we tested. Where they lead in each of these domains is for domain experts to determine. We share the toolkit. The applications are yours to discover.

20 Conclusion

We built two forward prediction engines for Riemann zeta zeros. We tested them on 80,000 validated zeros across ten orders of magnitude. We released the code, the parameters, the validation protocol, and the results. We documented what works, what is preliminary, and what is open.

ARE+ and Cross-Wells predict the next zero before it is observed. They do this in 1.2–2.5 microseconds on standard hardware. They maintain stability across regimes without retraining. They beat statistical baselines on every tested metric. This is what we found. It works today within current zeta structure understanding.

The PFSPC prime-counting framework connects zero predictions to prime distribution. The framework is correct. The benchmarks are pending. We share it as a foundation for others to build on and validate.

The spectral fingerprint candidate (0.977, 0.651) is an empirical observation we cannot yet explain theoretically. It persists across nine height regimes. It differs from all tested controls. It may mean something. It may not. Wider L-function testing will tell.

These equations are not perfect. They are not the final word on zeta zeros or primes. They are tools built by independent researchers with limited resources, honest about their limitations, and shared openly with anyone who wants to use them. If they help — in research, engineering, physics, or any domain we have not anticipated — that is enough.

v All code, parameters, and validation scripts are in Section 14 (Python Toolkit). True zero values: LMFDB at lmfdb.org and Odlyzko tables at dtc.umn.edu/~odlyzko/zeta_tables. Independent replication and improvement are welcomed.